

Problem Set 1

Applied Stats/Quant Methods 1

Due: October 8, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 8, 2026. No late assignments will be accepted.

Question 1: Education

A school counselor was curious about the average of IQ of the students in her school and took a random sample of 25 students' IQ scores. The following is the data set:

```
1 y <- c(105, 69, 86, 100, 82, 111, 104, 110, 87, 108, 87, 90, 94, 113, 112, 98,  
      80, 97, 95, 111, 114, 89, 95, 126, 98)
```

Number of Observations:

```
1 n = length(y)
```

Visualising the Data:

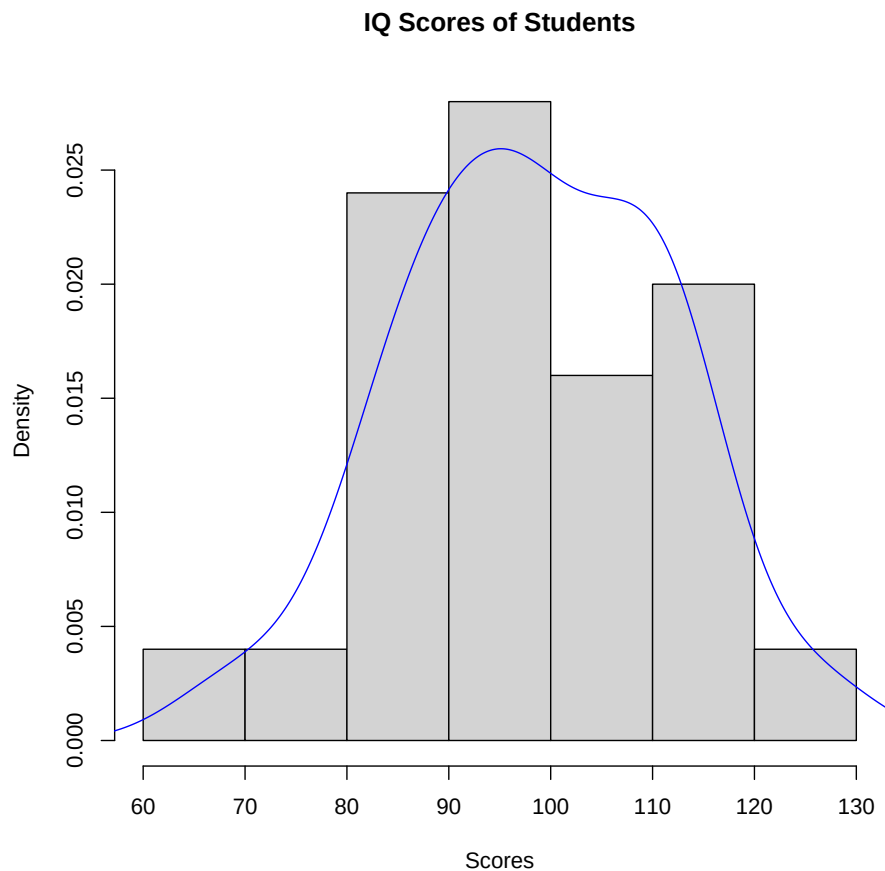
```
1 pdf("plot_1.pdf")  
2  
3 hist(y,  
4     breaks = 8,  
5     probability = TRUE,  
6     main = "IQ Scores of Students",  
7     xlab = "Scores")
```

```

8
9 lines(density(y), col = "blue")
10
11 dev.off()

```

Figure 1: IQ Scores of Students



1. Find a 90% confidence interval for the average student IQ in the school (As a hint, you first need the mean and SD to create a CI):

Using the same sample, conduct the appropriate hypothesis test with $\alpha = 0.05$.

Descriptive Statistics and Confidence Interval:

```

1 mean_y <- sum(y)/n
2 sd_y <- sqrt(sum((y - mean_y) ^ 2) / (n - 1))
3 se_y <- sd_y/sqrt(n)

```

```

4
5 # calculating the t-critical value: we use the t-distribution instead of
  the z-distribution because we observe that the length of y is less
  than 30 and the population mean and standard deviations are both
  unknown.
6 t_score <- qt(0.95, df = n - 1)
7
8 # calculating the lower and upper confidence bounds: confidence interval
  = sample statistic +/- margin of error
9 ci_90_lower = mean_y - t_score * se_y
10 ci_90_upper = mean_y + t_score * se_y

```

2. Next, the school counselor was curious whether the average student IQ in her school is higher than the average IQ score (100) among all the schools in the country.

Hypothesis Test Calculations:

```

1 mu_0 = 100
2
3 # iii. calculate a test statistic and t-critical value:
4 t_critical <- qt(0.95, df = n - 1)
5
6 t_score <- (mean_y - mu_0) / se_y
7
8 # iv. calculate a p-value
9 p_value <- pt(t_score, df = n-1, lower.tail = FALSE)

```

Check Using the Built-in `t.test()` Function:

```

1 t.test(y, mu = 100, alternative = "greater", conf.level = 0.95)

```

Question 2: Political Economy

Researchers are curious about what affects the amount of money communities spend on addressing homelessness. The following variables constitute our data set about social welfare expenditures in the USA.

State	50 states in US
Y	per capita expenditure on shelters/housing assistance in state
X1	per capita personal income in state
X2	Number of residents per 100,000 that are "financially insecure" in state
X3	Number of people per thousand residing in urban areas in state
Region	1=Northeast, 2= North Central, 3= South, 4=West

Explore the `expenditure` data set and import data into R.

```
1 expenditure <- read.table("https://raw.githubusercontent.com/ASDS-TCD/StatsI_2026/main/datasets/expenditure.txt", header=T)
```

- Please plot the relationships among Y, X1, X2, and X3? What are the correlations among them (you just need to describe the graph and the relationships among them)?

Scatterplots and Correlations:

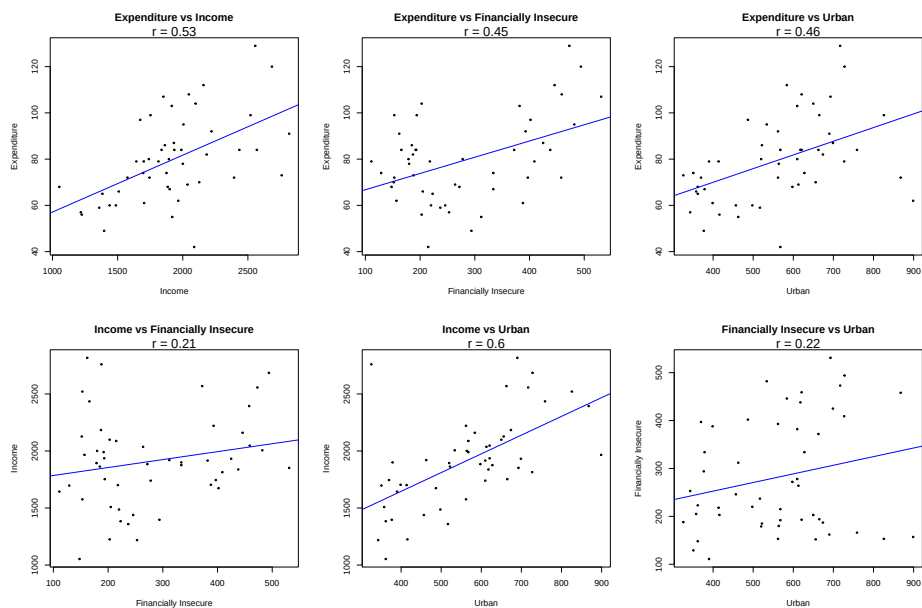
```
1 vars <- c("Y", "X1", "X2", "X3")
2
3 labels <- c(
4   "Expenditure",
5   "Income",
6   "Financially Insecure",
7   "Urban"
8 )
9
10 # use scatterplots for each panel because it most accurately represents
    the correlation between two continuous variables, displaying the
    magnitude and direction with the line of best fit
11 # scatterplots also efficiently visualise any outliers that may be
    skewing the data
12 pdf("plot_2.pdf", width = 12, height = 8)
13
14 par(mfrow = c(2, 3))
15
16 # use for loop to create the same six graphs with its respective
    variables
17 # abline creates a line of best fit
18 # mtext displays the correlation r value
19 for (i in 1:(length(vars) - 1)) {
20   for (j in (i + 1):length(vars)) {
```

```

21
22 plot(
23   expenditure[[vars[j]]],
24   expenditure[[vars[i]]],
25   xlab = labels[j],
26   ylab = labels[i],
27   main = paste(labels[i], " vs", labels[j]),
28   cex = 0.4
29 )
30
31 abline(
32   lm(expenditure[[vars[i]]] ~ expenditure[[vars[j]]]),
33   col = "blue",
34 )
35
36 r <- cor(
37   expenditure[[vars[i]]],
38   expenditure[[vars[j]]]
39 )
40
41 mtext(
42   paste("r =", round(r, 2)),
43
44 )
45 }
46 }
47
48 dev.off()

```

Figure 2: Pairwise Relationships and Correlations



- Please plot the relationship between Y and *Region*? On average, which region has the highest per capita expenditure on housing assistance?

Region Labels:

```
1 regions <- factor(
2   expenditure$Region,
3   levels = c(1, 2, 3, 4),
4   labels = c("North East", "North Central", "South", "West")
5 )
```

Comparison of Means and Medians:

```
1 for (i in levels(regions)) {
2   region_mean <- round(mean(expenditure$Y[regions == i]), 2)
3   region_median <- round(median(expenditure$Y[regions == i]), 2)
4
5   print(paste(i, "mean expenditure =", region_mean))
6   print(paste(i, "median expenditure =", region_median))
7   cat("\n")
8 }
```

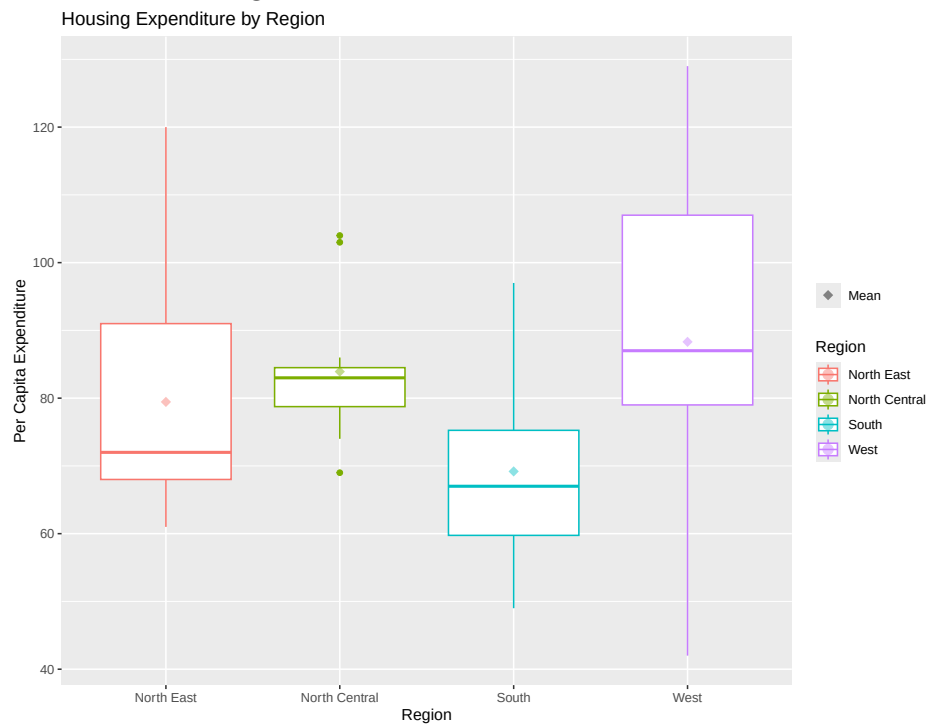
Housing Expenditure by Region:

```
1 plot_3 <- ggplot(expenditure, aes(x = regions, y = Y, color = factor(
2   regions))) +
3   geom_boxplot() +
4   stat_summary(
5     aes(shape = "Mean"),
6     fun = mean,
7     geom = "point",
8     size = 3,
9     alpha = 0.45
10  ) +
11  labs(
12    title = "Housing Expenditure by Region",
13    x = "Region",
14    y = "Per Capita Expenditure",
15    color = "Region",
16    shape = ""
17  ) +
18  scale_shape_manual(
19    values = c("Mean" = 18)
20  )
21 ggsave("plot_3.pdf", plot = plot_3, width = 9, height = 7)
```

- Please plot the relationship between Y and $X1$? Describe this graph and the relationship. Reproduce the above graph including one more variable *Region* and display different regions with different types of symbols and colors.

Housing Expenditure and Income:

Figure 3: Housing Expenditure by Region



```

1 plot_4 <- ggplot(expenditure, aes(x = X1, y = Y)) +
2   geom_point() +
3   geom_smooth(
4     method = "lm",
5     color = "blue",
6     se = FALSE
7   ) +
8   labs(
9     title = "The Relationship between Housing Expenditure and Income by
10    Region",
11    subtitle = paste("r =", round(cor(expenditure$X1, expenditure$Y), 3))
12    ,
13    x = "Per Capita Income",
14    y = "Per Capita Expenditure"
15  )
16 ggsave("plot_4.pdf", plot = plot_4, width = 9, height = 7)

```

Housing Expenditure and Income by Region:

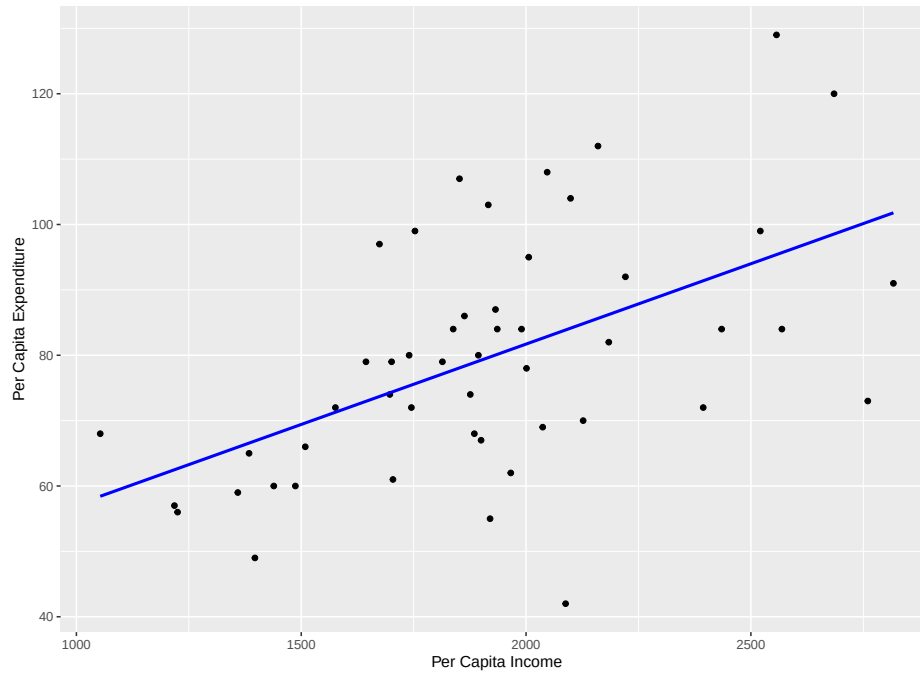
```

1 plot_5 <- ggplot(
2   expenditure,
3   aes(
4     x = X1,
5     y = Y,
6     color = factor(regions),

```

Figure 4: Housing Expenditure versus Income

The Relationship between Housing Expenditure and Income by Region
 $r = 0.532$



```

7   shape = factor(regions)
8   )
9   ) +
10  geom_point() +
11  labs(
12    title = "The Relationship between Housing Expenditure and Income by
13    Region",
14    x = "Per Capita Income",
15    y = "Per Capita Expenditure",
16    color = "Region",
17    shape = "Region"
18  )
19  ggsave("plot_5.pdf", plot = plot_5, width = 9, height = 7)

```

Correlations by Region:

```

1  for (i in levels(regions)) {
2    region_correlation <- round(cor(expenditure$X1[regions == i],
3    expenditure$Y[regions == i]), 3)
4
5    print(paste(i, "correlation =", region_correlation))
6    cat("\n")
7  }

```

Figure 5: Housing Expenditure versus Income by Region
The Relationship between Housing Expenditure and Income by Region

